

Summary

We propose two practical e-variable classes for implementing sequential conditional independence tests under the **Model-X** assumption.

Problem statement

Ingredients:

- A (partially) trained ML model:

$$Y \approx \hat{m}(X, Z),$$

optimized for minimizing a **loss function** ℓ , with a **feature of interest** X .

- True conditional distribution** $P_{X|Z}^*$ (MX assumption) – or good estimate.
- Can draw samples $\tilde{X} \sim P_{X|Z}^*(z)$ for any z .

Main goal: sequential test for null hypothesis

$$\mathcal{H}_{P_{X|Z}^*}^{\text{MX}} := \{P(X, Y, Z) : Y \perp_P X|Z, P_{X|Z} = P_{X|Z}^*\},$$

which:

- Is **anytime-valid** whenever $P_{X|Z}^*$ is known exactly.
- Is **powerful** under a wide range of alternatives.
- Provides approximate type-1 error guarantees when $P_{X|Z}^*$ is estimated.
- Can make use of any **machine learning** model $\hat{m}$.

The GRO e-variable [Grünwald et al., 2024]

The **GRO e-variable** for testing $\mathcal{H}_{P_{X|Z}^*}^{\text{MX}}$ vs $\{Q\}$ is:

$$\frac{f_{Y|X,Z}(Y|X, Z)}{f_{Y|Z}(Y|Z)} = \frac{f_{Y|X,Z}(Y|X, Z)}{\mathbb{E}_{\tilde{X}|Z}[f_{Y|X,Z}(Y|\tilde{X}, Z)]},$$

where Q has density $f(x, y, z)$ and $\tilde{X} \sim P_{X|Z}^*(Z)$.

Practical limitation: Hard to estimate $f_{Y|X,Z}$.

Antisymmetric e-variables [Shaer et al., 2023]

To incorporate machine learning models, use **importance statistics**:

$$l := \ell(\hat{m}(X, Z), Y), \quad \tilde{l} := \ell(\hat{m}(\tilde{X}, Z), Y),$$

and e-variables of the form:

$$1 + g(l, \tilde{l}),$$

with **antisymmetric** functions $g(g(a, b) = -g(b, a))$ such that $g(a, b) \geq -1$.

- Simplifies conditional independence question into **exchangeability**

$$Y \perp X|Z \longrightarrow (l, \tilde{l}) \stackrel{d}{=} (\tilde{l}, l).$$

- Examples:

$$1 + \lambda \text{sign}(\tilde{l} - l), \quad 1 + \lambda \tanh(\tilde{l} - l), \quad 1 + \lambda(\tilde{l} - l).$$

Practical limitation: Function g is left unspecified, unclear how to choose it.

Candidate I: Exponential e-variable

$$E_{\text{exp}}^\eta := \frac{e^{-\eta l}}{\mathbb{E}_{\tilde{X}|Z}[e^{-\eta \tilde{l}}]},$$

with a free parameter $\eta \geq 0$.

- Good **approximation of the GRO** e-variable whenever:

$$\ell(m(x, z), y) \approx -\eta \log(f_{Y|X,Z}(y|x, z)).$$

- Contains GRO in regression tasks with squared loss and Gaussian noise, and classification with cross-entropy.

Candidate II: Antisymmetric KDE e-variable

$$E_{\text{KDE}}^{\hat{f}} := 1 + \mathbb{E}_{\tilde{X}|Z} \left[\frac{\hat{f}(l, \tilde{l}) - \hat{f}(\tilde{l}, l)}{\hat{f}(l, \tilde{l}) + \hat{f}(\tilde{l}, l)} \right] = \mathbb{E}_{\tilde{X}|Z} \left[\frac{2\hat{f}(l, \tilde{l})}{\hat{f}(l, \tilde{l}) + \hat{f}(\tilde{l}, l)} \right],$$

where $\hat{f}$ approximates the density of $(l, \tilde{l})$ via Kernel Density Estimation.

- The GRO e-variable for testing exchangeability null $(A, B) \stackrel{d}{=} (B, A)$ against $Q(A, B)$ with density $f(a, b)$ is:

$$\frac{2f(A, B)}{f(A, B) + f(B, A)}.$$

- For good approximations $\tilde{f} \approx f$, for any antisymmetric g :

$$\mathbb{E} \left[\log \left(\frac{2\hat{f}(l, \tilde{l})}{\hat{f}(l, \tilde{l}) + \hat{f}(\tilde{l}, l)} \right) \right] \geq \mathbb{E} \left[\log (1 + g(l, \tilde{l})) \right].$$

- We simplify the null to $(l, \tilde{l}) \stackrel{d}{=} (\tilde{l}, l) + \text{Model-X assumption}$.

- $E_{\text{KDE}}^{\hat{f}}$ approximates the GRO **among antisymmetric** e-variables!

→ **Approximate the antisymmetric GRO** with KDE and derandomize $\tilde{X}$.

Exponential vs antisymmetric e-power

- Whenever $\ell(m(x, z), y) = -\eta \log(f_{Y|X,Z}(y|x, z))$:
 E_{exp}^η is GRO $\Rightarrow$ dominates all antisymmetric e-variables.
- Do antisymmetric e-variables ever dominate?
- Yes! E.g., the **coin betting** e-variable (defined for bounded ℓ):

$$E_{\text{cb}}^\lambda = 1 + \lambda(\tilde{l} - l).$$

can outperform the exponential in **small signal regimes**.

Thm: Exponential vs Coin betting e-power

Define *predictiveness signal*:

$$\Delta := \mathbb{E} [\ell(m(\tilde{X}, Z), Y)] - \mathbb{E} [\ell(m(X, Z), Y)] \geq 0,$$

and loss variances:

$$\sigma_0^2 := \mathbb{V}(\ell(m(\tilde{X}, Z), Y)), \quad \sigma_1^2 := \mathbb{V}(\ell(m(X, Z), Y)).$$

Optimal parameters and corresponding e-powers for E_{exp}^η and E_{cb}^λ are:

$$\eta^* = \frac{\Delta}{\sigma_0^2} + O(\Delta^2), \quad \lambda^* = \frac{\Delta}{\sigma_1^2 + \Delta^2} + O(\Delta^2)$$

$$\mathbb{E} [\log E_{\text{exp}}^{\eta^*}] = \frac{\Delta^2}{2\sigma_0^2} + O(\Delta^3), \quad \mathbb{E} [\log E_{\text{cb}}^{\lambda^*}] = \frac{1}{2} \frac{\Delta^2}{\sigma_1^2 + \Delta^2} + O(\Delta^3).$$

Takeaway:

- In small signal regimes, the e-power ratio is: $\frac{\mathbb{E} [\log E_{\text{cb}}^{\lambda^*}]}{\mathbb{E} [\log E_{\text{exp}}^{\eta^*}]} \approx \frac{\sigma_0^2}{\sigma_1^2 + \Delta^2}$.

→ **Coin betting can be more e-powerful!**

Experiments

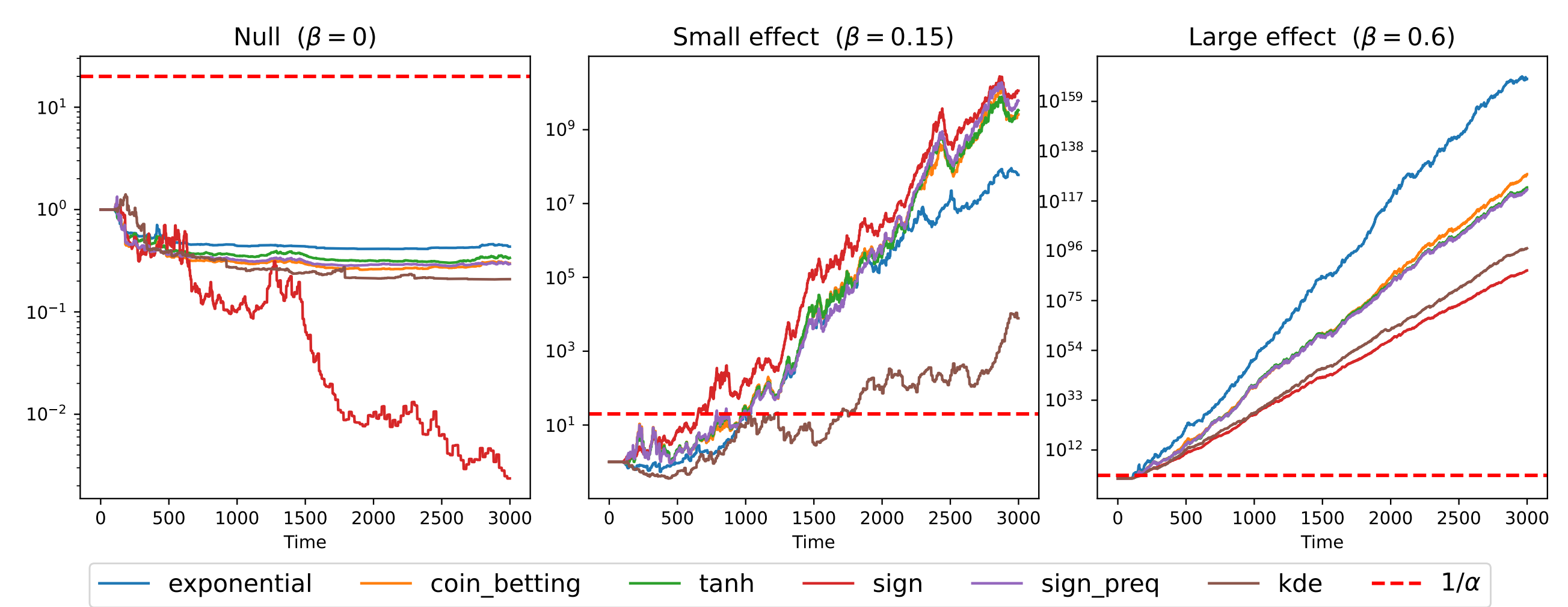


Figure: **Simulated data**. $y = (Z^\top W)^2 + \beta X + \varepsilon$, where $X | Z \sim \mathcal{N}(U^\top Z, 1)$ and $Z \sim \mathcal{N}(0, I_d)$ with $d = 19$. Vectors U and W are sampled uniformly from the simplex. Results are averaged over 20 random seeds and $\hat{m}$ is a Gradient Boosting.

- Under the null ($\beta = 0$): neither process rejects.
- For a weak signal ($\beta = 0.15$): antisymmetric e-variables beat the exponential.
- For a stronger signal ($\beta = 0.6$): the exponential exhibits fastest growth.

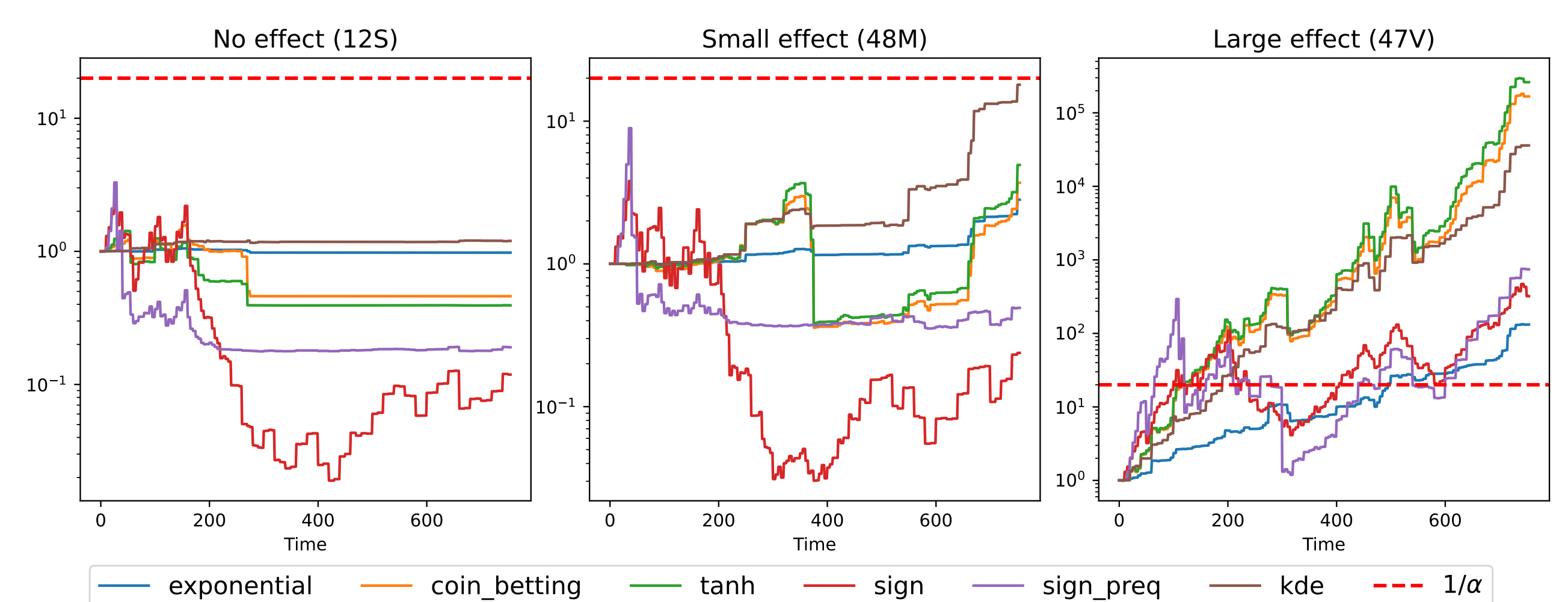


Figure: **HIV data**. ML e-processes for detecting mutations in HIV associated with drug resistance using a Lasso regressor.

- Antisymmetric may be preferable with real-world misspecifications.

Conclusions

Machine Learning can effectively be used for Sequential Scientific Discovery!

- We make the GRO e-variable practical through the exponential e-variable.
- Under misspecification, antisymmetric e-variables can dominate.
- We propose the GRO antisymmetric based on estimating a 2-d density.
- Ongoing work:** we are studying:
 - The KDE e-variable and the error induced by estimating density $\hat{f}$.
 - Safety guarantees when the Model-X distribution $P_{X|Z}^*$ is itself estimated.

References

- P. Grünwald, A. Henzi, and T. Lardy. Anytime-valid tests of conditional independence under model-x. JASA, 2024.
- S. Shaer, G. Maman, and Y. Romano. Model-x sequential testing for conditional independence via testing by betting. In AISTATS, 2023.